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Let’s learn about **satellites**!

Space mission and **satellite** design course

Satellite and space missions

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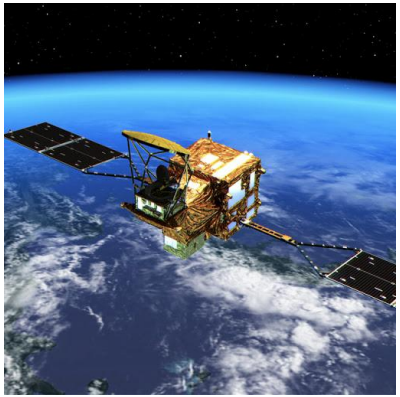
What is satellite?

“A satellite or artificial satellite is an object intentionally placed into orbit around a celestial body”.



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What is satellite?

- In its most common form, a satellite is a semi-autonomous system designed to carry out tasks and operations in orbit.
- It is launched into space and inserted into an orbit around a celestial body, most often Earth.
- A satellite doesn't "fly" like an aircraft: it follows a gravitational trajectory, typically an elliptical orbit with the central body located at one of the two foci.

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What is satellite?

- Most satellites are characterized by a key system called the **PAYLOAD**.
- The payload is responsible for performing the mission's core function: the operation or service that justifies the satellite's design, construction, and launch.
- The rest of the satellite, which supports and enables the payload, is commonly referred to as the **BUS**.

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What is a satellite?

- By placing a satellite in orbit and performing operations we realize a **space mission**.
- A space mission comprehend all the activities from the conception of the satellite up to the end of operation.
- Each space mission is characterized by 3 relevant segments

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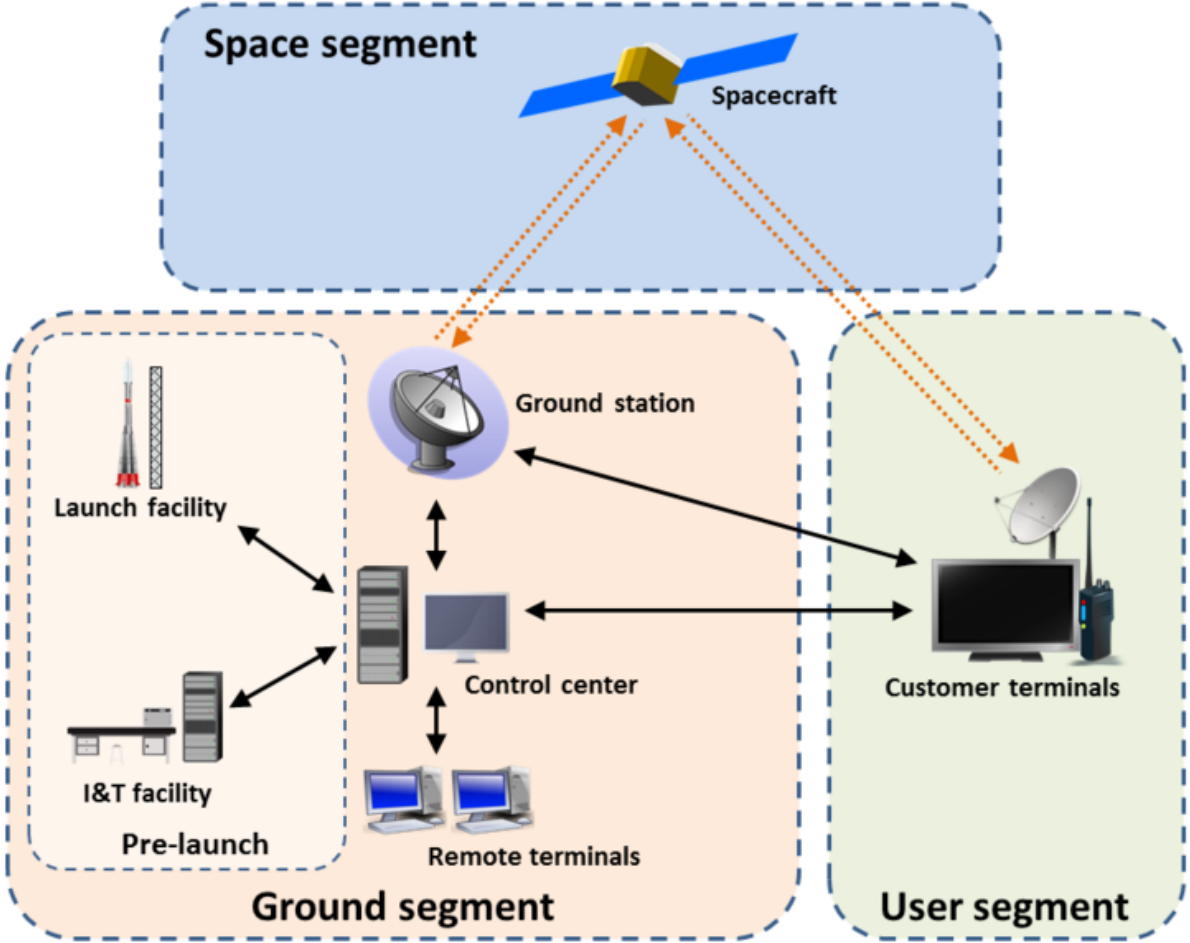

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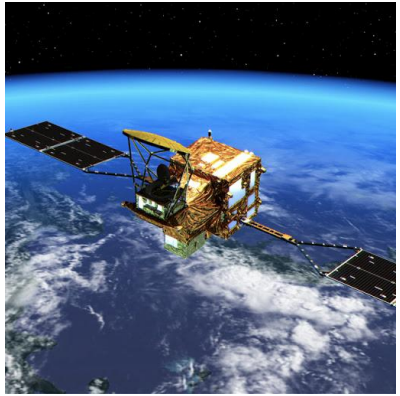
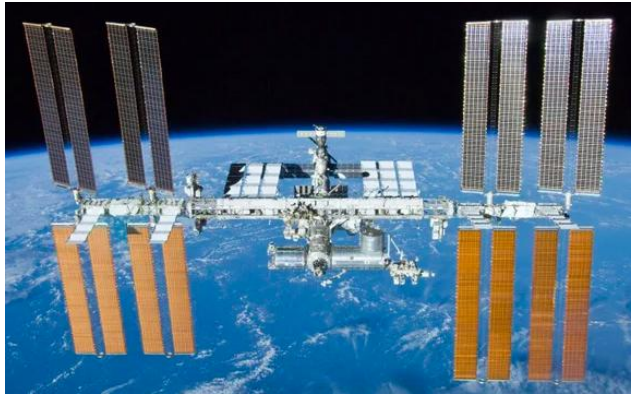
Why do we launch satellites?

- A satellite is constructed and placed in orbit in order to perform several operation aimed to support **human life** on earth and beyond.





Question: what are these satellites doing for human life on earth?

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Why do we launch satellites?

Every satellite starts from a need: a problem we want to solve or a service we want to provide.

Scope

The mission objective, the problem to solve, the data to collect, or the service to deliver.

Payload

The instrument or system that performs the mission and enables the service.

Satellite

The space platform that brings the payload into orbit and provides power, control, communication, and operation

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Satellite design

- The **design of the satellite** starts from definition of the **payload** requirements and constraints.

- The **payload design** starts from the definition of **mission scope** requirements and constraints

- The **mission** design starts from the understanding of the **problem** i need to solve and the way to pursue that (the data to be collected, the operation to be performed)

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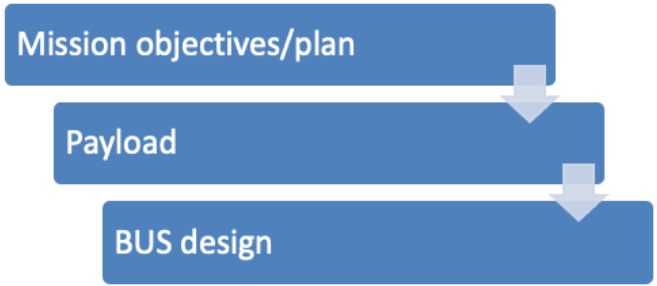
Satellite design

Why are these critical steps?

Because mistakes made in the early stages create significant complications later on, and misjudging payload requirements leads to exponentially increasing costs and constraints throughout design, integration, launch, and operations.



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Satellite classification

- **Communications**: make the connection through radio, television, and telephones.
- **Weather**: used for giving out continuous or up-dated pieces of information concerning the atmospheric conditions.
- **Research**: used for understanding the basic properties of materials and process in space, and to explore outer space, including features of celestial objects, magnetic fields, etc.
- **Application**: satellites used and designed to examine and test ways for improving the satellites technology of outer space.
- **Navigational**: satellites developed for navigation on earth
- **Earth observation**: satellite to perform remote sensing of earth
- **Space observation**: satellite to perform to perform space objet traffic management (space observation)

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Satellite classification

Satellites are normally classified also in terms of mass:

- Large satellites: More than 1,000 kg.
- Medium-sized satellites: 500-1,000 kg.
- Minisatellite: 100-500 kg.
- **Microsatellite: 10-100 kg.**
- **Nanosatellite: 1-10 kg.**
- Picosatellite: Less than 1 kg.

SmallSats

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CubeSat

The CubeSat standard was introduced in 1999 by Prof. Jordi Puig-Suari at California Polytechnic State University (Cal Poly) and Prof. Bob Twiggs at Stanford University.

CubeSat

- A CubeSat is a standardized small satellite based on a 1U unit measuring 10 × 10 × 10 cm, typically with a mass of up to about 1.33 kg per 1U.

- CubeSats scale by combining multiple units (e.g., 2U, 3U, 6U, 12U, 16U), while keeping the same underlying form factor.

- Regardless of size, CubeSats follow common mechanical and deployment interfaces and standardized layouts defined in the CubeSat Design Specification.

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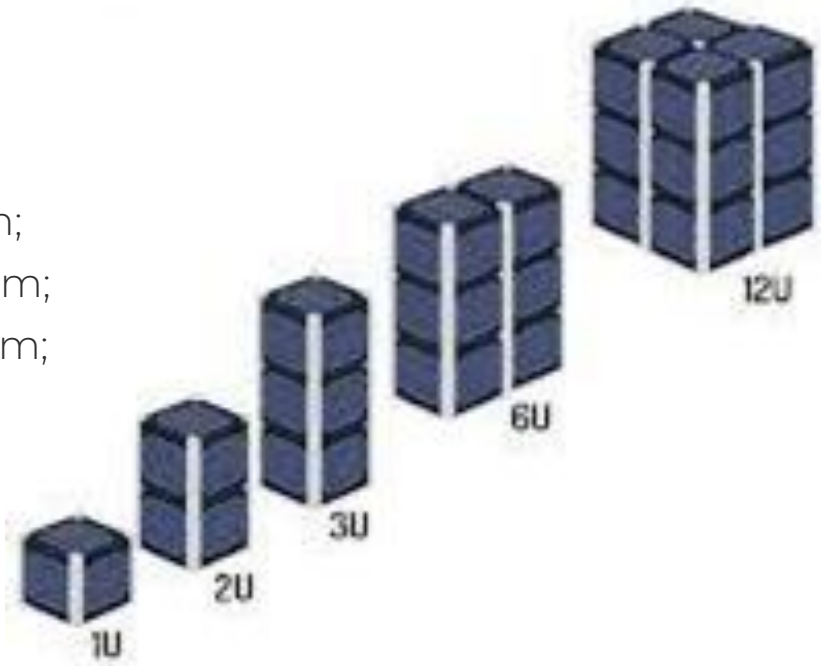
CubeSat

- 1U: 2 Kg - volume 100x100x100mm;
- 2U: 4 Kg - volume 100x100x227mm;
- 3U: 6 kg - volume 100x100x340mm;
- 6U: 12 Kg - volume 200x100x340mm;
- 12U: 24 kg - volume 200x200x340mm;
- 16U: 32 kg - volume 200x200x454mm;



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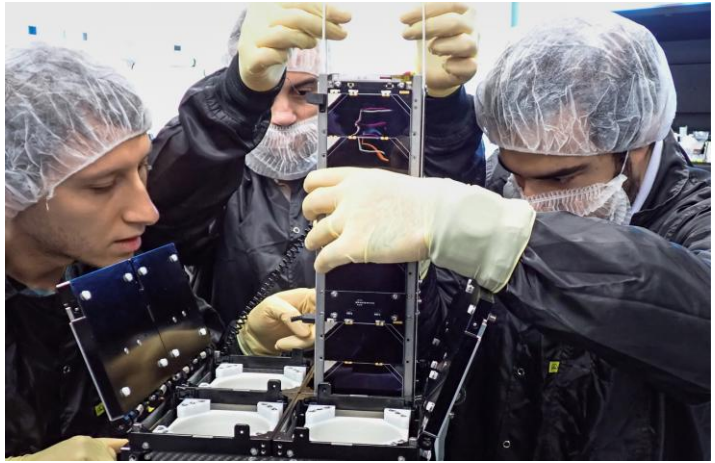

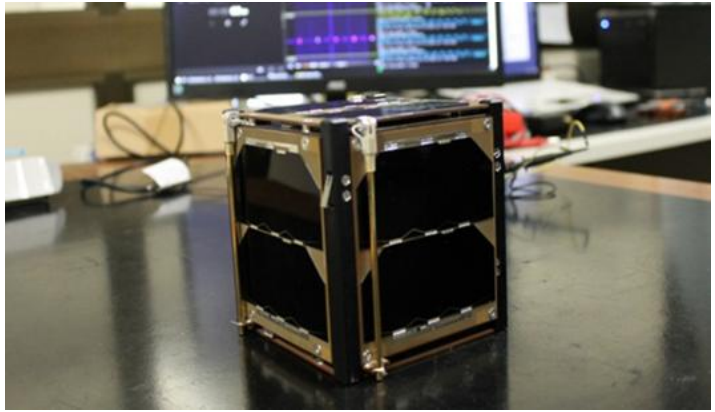

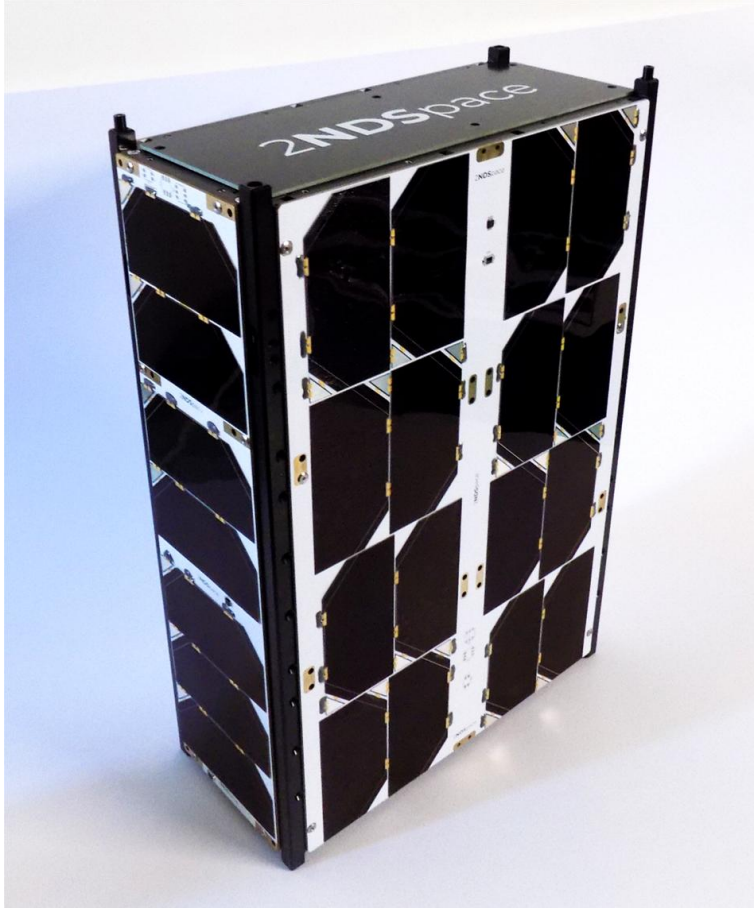

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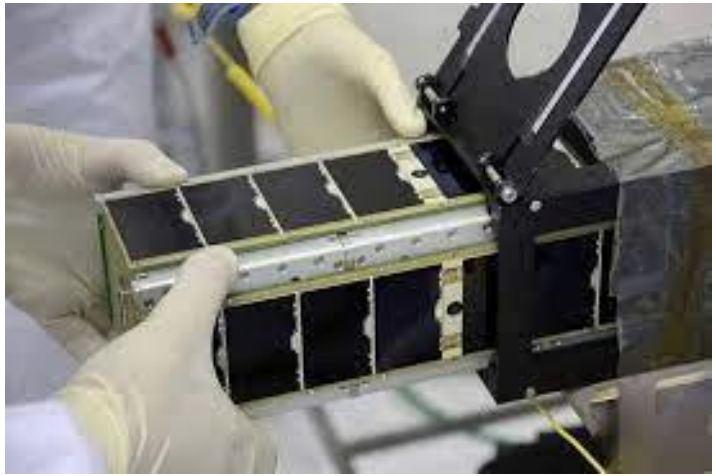
CubeSat

- The CubeSat standard introduced standardization in the deployment system - the device that allows the satellite to safely separate from the launch vehicle once in orbit.

- This system, called the deployer, releases and ejects the satellite into space in a controlled manner.



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CubeSat

Like Cubesats, deployers are characterized by different sizes

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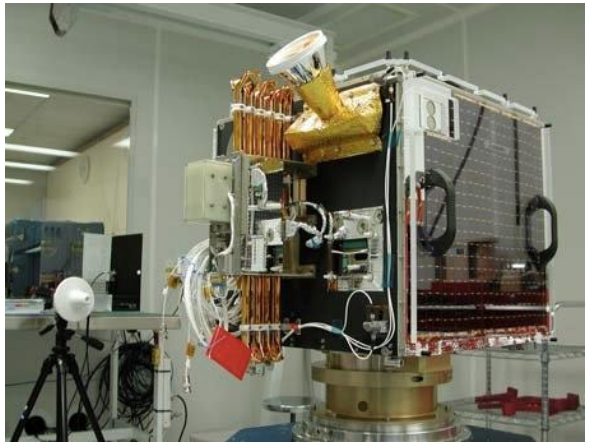
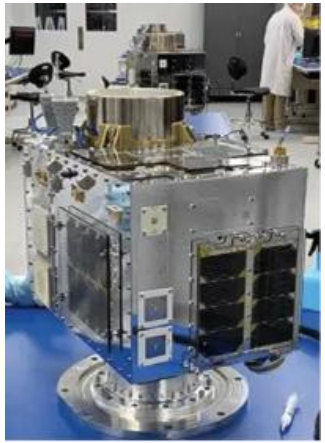
Spot the difference - 1





https://www.researchgate.net/figure/The-FedSat-satellite-during-integration_fig3_237643943

https://www.utias-sfl.net/what-we-do/satellite-development-overview/

https://spacenews.com/atk-ships-ors-1-bus-prime-contractor/

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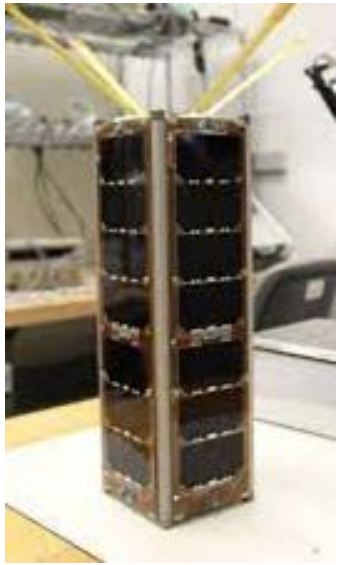
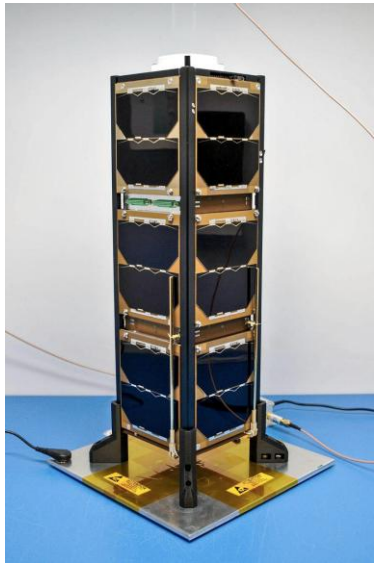
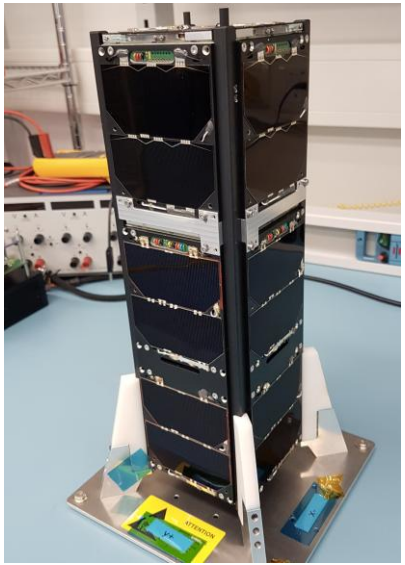
Spot the difference - 2







https://www.nasa.gov/feature/elana-31-mission-cubesats-deployed

https://commons.wikimedia.org/wiki/File:The_RAX-2_Spacecraft.jpg

https://spaceflight101.com/pslv-c37/pslv-c37-cubesats/

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CubeSat

- By introducing standardized geometry and interfaces, companies gained the ability to develop **subsystems** based on shared and well-defined requirements.

- This standardization enabled the creation of a true **market** ecosystem, with the emergence of specialized subsystem manufacturers, platform providers, and service companies.

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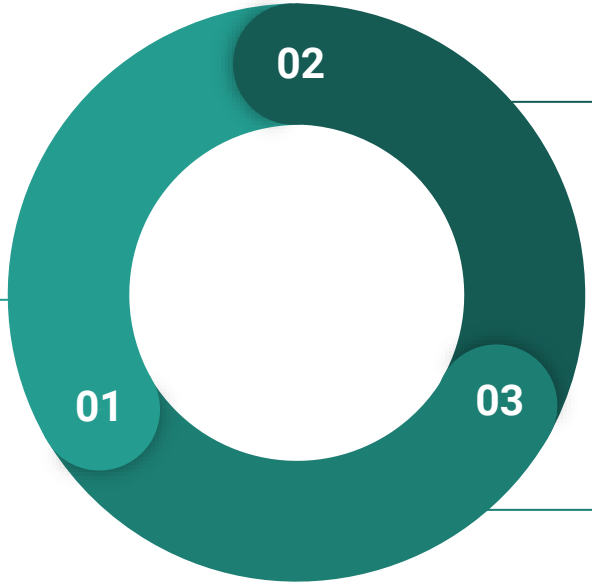
CubeSat - Advantages

Lower cost and faster development

Standardized dimensions and interfaces make CubeSats easier to design, integrate, and launch, reducing time and overall mission cost.



Technology became smaller and more powerful

The miniaturization of electronics and the availability of high-performance commercial components made it possible to build capable satellites in very small volumes.

The space market expanded

Growing demand for space-based data and services, along with increased private investment, accelerated the adoption of CubeSats.

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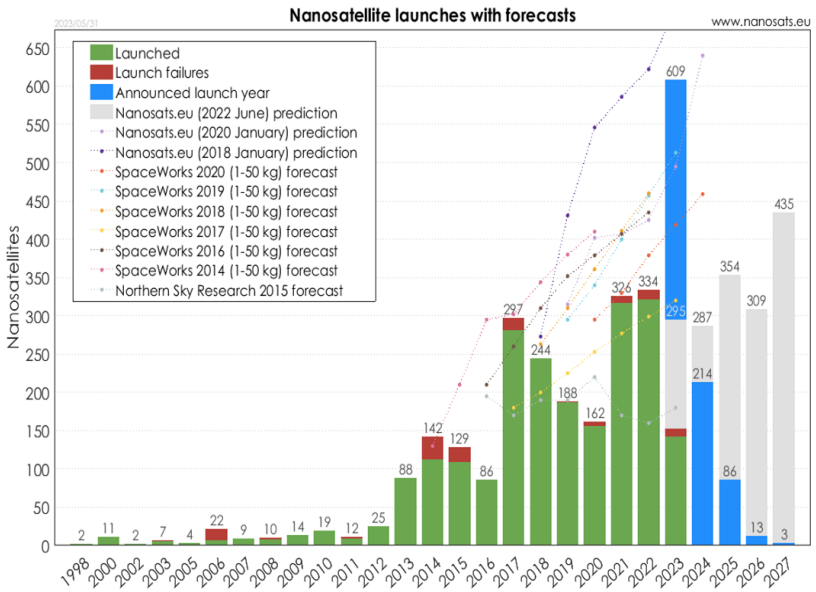
CubeSat



- Nanosatellite launches have grown exponentially since 1998.
- Over 3,200 nanosats have been launched as of early 2026.
- Future forecasts predict tens of thousands of small satellites deployed over the next decade.

https://www.nanosats.eu/

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